

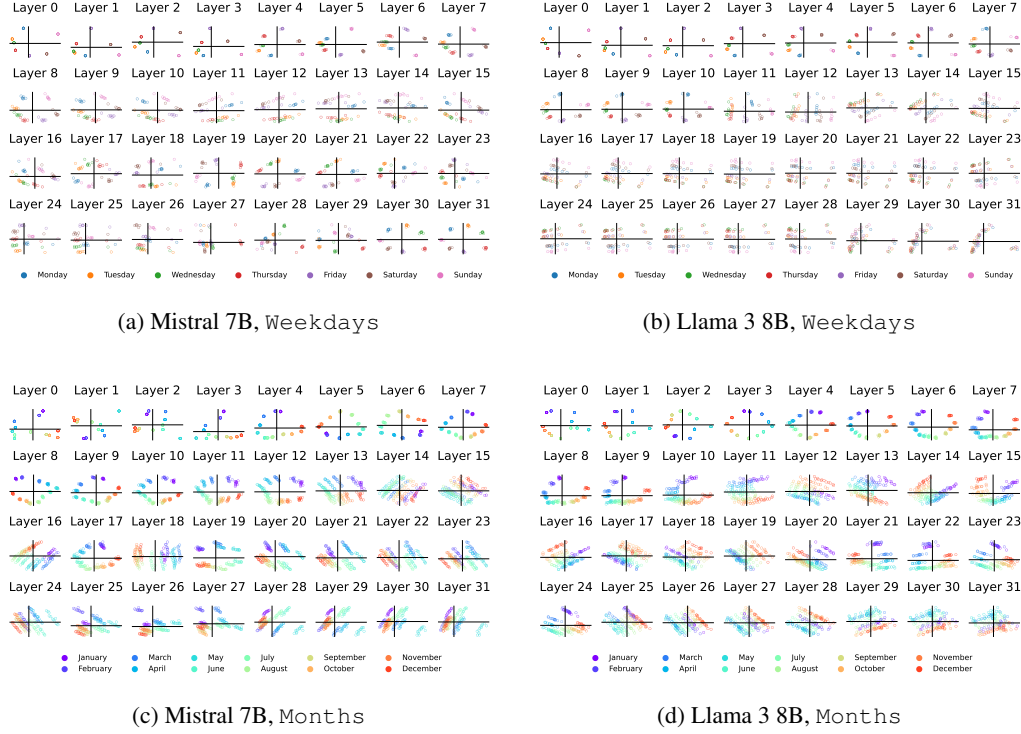


Figure 18: Projections onto the top two PCA dimensions of model hidden states on the α token show that circular representations of α are present in various layers.

The reason that the Months error bars are smaller than the Weekdays error bars is because there are more Months prompts: there are $12 * 12 * 11 = 1584$ intervention effect values, rather than $7 * 7 * 6 = 294$ intervention effect values.

I MORE WEEKDAYS AND MONTHS PLOTS AND DETAILS

I.0.1 BASIC PLOTS

We show the results of Mistral 7B and Llama 3 8B on all individual instances of Weekdays that at least one of the models get wrong in Table 2 and present a similar table for Months in Table 3.

We show projections onto the top two PCA directions for both Mistral 7B and Llama 3 8B in Fig. 18 on the hidden layers on top of the α token, colored by α . These are similar plots to Fig. 4, except they are on all layers. The circular structure in α is visible on many—but not all—layers. Much of the linear structure visible is due to β .

I.0.2 INTERVENING WITH THE SAE PROBE

We show the results of projecting Mistral Weekdays representations into the plane discovered by clustering SAE features in Fig. 19. The result is clearly circular.

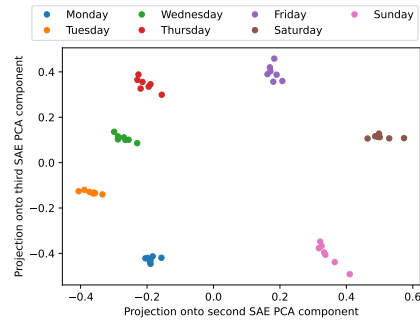


Figure 19: Projections of Mistral 7B Weekdays task activations at layer 8 into the plane discovered by clustering layer 8 SAE features.

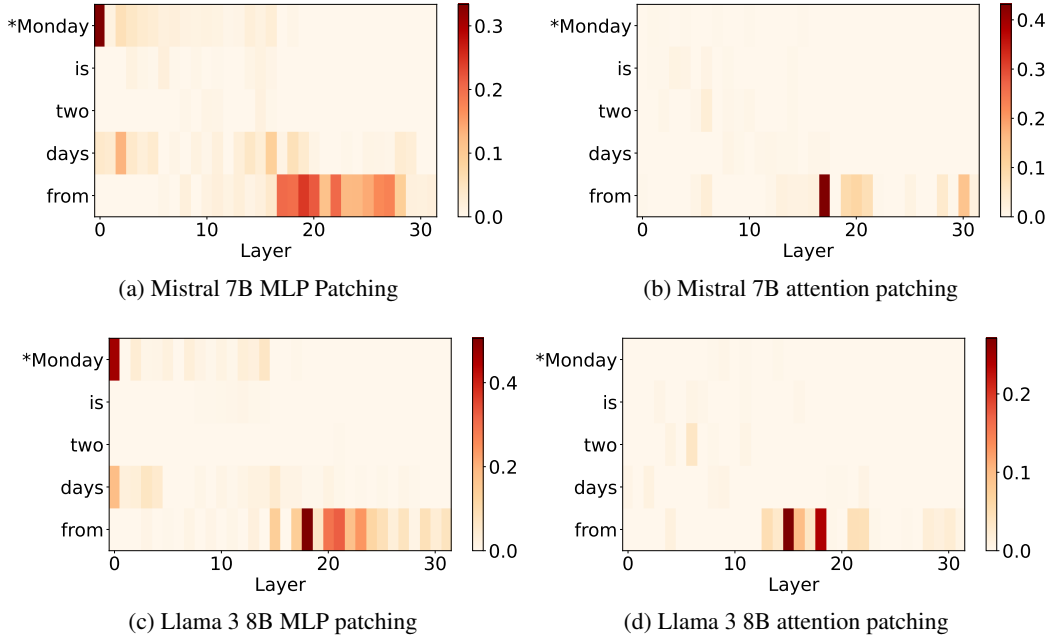


Figure 20: Attention and MLP patching results on Weekdays. Results are averaged over 20 different runs with fixed α and varying β and 20 different runs with fixed β and varying α .

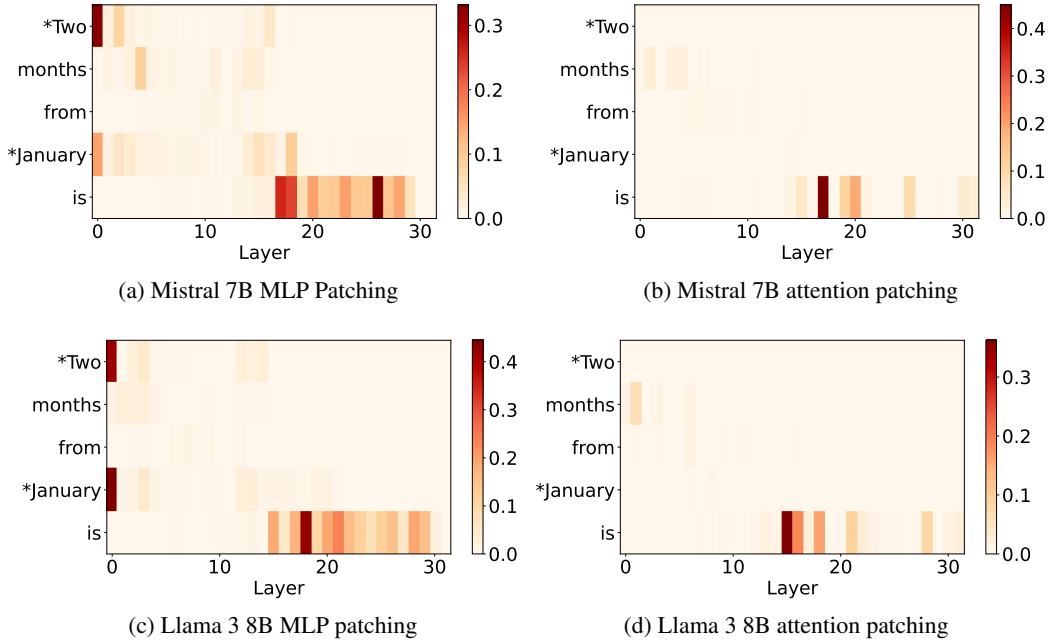


Figure 21: Attention and MLP patching results on Months. Results are averaged over 20 different runs with fixed α and varying β and 20 different runs with fixed β and varying α .

I.0.3 BASIC PATCHING

In Fig. 20 and Fig. 21, we report MLP and attention head patching results for Weekdays and Months. We experiment on 20 pairs of problems with the same α and different β and 20 pairs of problems with the same β and different α , for a total of 40 pairs of problems. For each pair

Table 2: Weekdays finegrained results. Row omitted if both models get it correct.

α	β	Ground truth γ	Mistral top γ	Mistral correct?	Llama top γ	Llama correct?
1	1	Wednesday	Wednesday	Yes	Thursday	No
3	1	Friday	Friday	Yes	Tuesday	No
4	1	Saturday	Saturday	Yes	Thursday	No
3	2	Saturday	Saturday	Yes	Tuesday	No
4	2	Sunday	Sunday	Yes	Wednesday	No
5	2	Monday	Monday	Yes	Tuesday	No
2	3	Saturday	Friday	No	Saturday	Yes
3	3	Sunday	Sunday	Yes	Tuesday	No
4	3	Monday	Monday	Yes	Tuesday	No
0	4	Friday	Thursday	No	Friday	Yes
3	4	Monday	Monday	Yes	Tuesday	No
0	5	Saturday	Friday	No	Saturday	Yes
1	5	Sunday	Saturday	No	Wednesday	No
2	5	Monday	Sunday	No	Monday	Yes
4	5	Wednesday	Tuesday	No	Tuesday	No
6	5	Friday	Thursday	No	Thursday	No
1	6	Monday	Sunday	No	Thursday	No
2	6	Tuesday	Monday	No	Tuesday	Yes
3	6	Wednesday	Tuesday	No	Tuesday	No
4	6	Thursday	Thursday	Yes	Tuesday	No
5	6	Friday	Friday	Yes	Thursday	No
6	6	Saturday	Thursday	No	Thursday	No
0	7	Monday	Sunday	No	Tuesday	No
1	7	Tuesday	Sunday	No	Tuesday	Yes
2	7	Wednesday	Sunday	No	Wednesday	Yes
3	7	Thursday	Sunday	No	Thursday	Yes
4	7	Friday	Thursday	No	Tuesday	No
5	7	Saturday	Friday	No	Saturday	Yes
6	7	Sunday	Friday	No	Thursday	No

of problems, we patch the MLP/attention outputs from the "clean" to the "dirty" problem for each layer and token, and then complete the forward pass. Defining the logit difference as the logit of the clean γ minus the logit of the dirty γ , we record what percent of the difference between the original logit difference of the dirty problem and the logit difference of the clean problem is recovered upon intervening, and average across these 40 percentages for each layer and token. This gives us a score we call the *Average Intervention Effect*.

For simplicity of presentation, we clip all of the (few) negative intervention averages to 0 (prior work (Zhang & Nanda, 2023) has also found negative-effect attention heads during patching experiments).

I.0.4 CIRCLE CONTINUITY

Finally, in Fig. 22, we show another example of the continuity of the circular days of the week representation in Mistral 7B.

J PATCHING

In this section, we present results to support a claim that MLPs (and not attention blocks) are responsible for computing γ . In Fig. 25, we deconstruct states on top of the final token (before predicting γ) on Llama 3 8B Months (we show a similar plot for the states on the final token of Mistral 7B on Weekdays in the main text in Fig. 24). These plots show that the value of γ is computed on the final token around layers 20 to 25. To show that this computation occurs in the MLPs, we must show that no attention head is copying γ from a prior token or directly computing γ .

Table 3: Months finegrained results. Row omitted if both models get it correct.

α	β	Ground truth γ	Mistral top γ	Mistral correct?	Llama top γ	Llama correct?
0	4	May	April	No	May	Yes
6	4	November	October	No	November	Yes
0	6	July	June	No	July	Yes
0	7	August	July	No	August	Yes
1	7	September	October	No	September	Yes
3	7	November	October	No	November	Yes
5	7	January	December	No	January	Yes
6	7	February	January	No	February	Yes
7	7	March	February	No	March	Yes
9	7	May	April	No	May	Yes
4	9	February	February	Yes	January	No
2	10	January	December	No	January	Yes
8	10	July	June	No	July	Yes
1	11	January	December	No	January	Yes
2	11	February	December	No	February	Yes
3	11	March	February	No	March	Yes
7	11	July	June	No	July	Yes
8	11	August	July	No	August	Yes
9	11	September	August	No	September	Yes
0	12	January	December	No	January	Yes

We first perform a patching experiment with the same setup Fig. 21 and Fig. 20 on individual attention heads on the final token. From the patching results we identify the top 10 attention heads by average intervention effect. For each attention head, we compute one EVR run with explanatory functions equal to one-hot functions of α and β (resulting in 14 functions g_i for Weekdays and 24 for Months) and one with explanatory functions equal to one-hot functions of α , β , and γ . We find that for all layers before 25, adding γ to the explanatory functions adds almost no explanatory power. Since we established above that the model has already computed γ at this point, we know that attention heads do not participate in computing γ .

To isolate the rough circuit for Weekdays and Months, we perform layer-wise activation patching on 40 random pairs of prompts. The results, displayed in Fig. 21 show that the circuit to compute γ consists of MLPs on top of the α and β tokens, a copy to the token before γ , and further MLPs there (roughly similar to prior work studying arithmetic circuits (Stolfo et al., 2023)). Moreover, fine-grained patching in Appendix K shows that there are just a few responsible attention heads for the writes to the token before γ . However, patching alone cannot tell use *how* or *where* γ is represented. For that, we need a new technique, which we expand on in the next section.

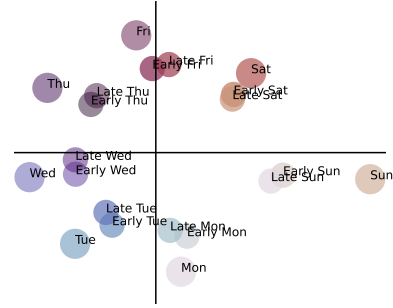


Figure 22: Layer 30 Mistral 7B activations for [very early/very late] on [Monday/Tuesday/.../Sunday], plotted projected into the PCA plane for [Monday/Tuesday/.../Sunday].

K EXPLANATION VIA REGRESSION (EVR)

So far, we have focused on examining and intervening on the representation for α , which we present as a circle in the top PCA components on top of the α token. In this section, we examine how the generated output, γ , is represented.

First, to isolate the rough circuit for Weekdays and Months, we perform layer-wise activation patching on 40 random pairs of prompts. The results, displayed in Fig. 21 and Fig. 20, show that the circuit to compute γ consists of MLPs on top of the α and β tokens, a copy to the token before

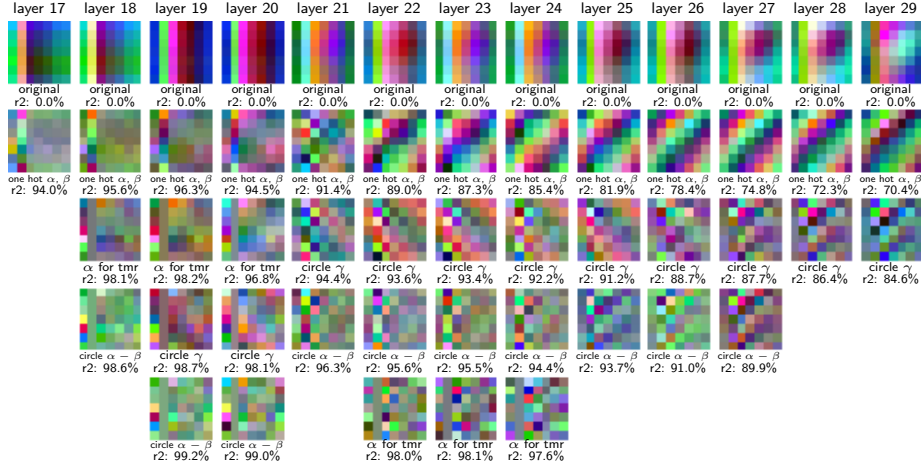


Figure 24: EVR residual RGB plots on Mistral hidden states on the `Weekdays` final token, layers 17 to 29. From top to bottom, we show each residual RGB plot after adding the function(s) g_i labelled just underneath, as well as the resulting r^2 value. We write “tmr” meaning “tomorrow” for $\beta = 1$. We also write “circle for x ” meaning the inclusion of two functions $g_i(x) = \{\cos, \sin\}(2\pi x/7)$.

γ , and further MLPs there (roughly similar to what Stolfo et al. (2023) find in prior work studying arithmetic circuits). Thus, we know *where* to look for a representation of γ : in the second half of the layers on the token before γ . However, patching alone cannot tell use *how* γ is represented.

Unlike α , γ has no obvious circular (or linear) pattern in the top PCA components on these layers. To determine the representation for γ , we introduce a more powerful technique we call Explanation via Regression (EVR): given a set of token sequences with a corresponding set of hidden states $X_{i,l}$, we choose a set of interpretable explanation functions of the input tokens $\{g_j(t)\}$. The r^2 value of a linear regression from $\{g_j(t)\}$ to $X_{i,l}$ tells us how much of the variance in the activations the $\{g_j(t)\}$ explain, and conversely the residuals show the exact components of the representation we have yet to explain.

K.1 USING EVR TO UNCOVER A CIRCULAR REPRESENTATION FOR γ

We first use EVR to determine the representation for γ by plotting the top two PCA components of the layer 25 Mistral 7B activations after subtracting the components that can be explained using a regression with one hot functions in α and β (i.e. $g_1 = [\alpha = 0]$, $g_2 = [\beta = 1]$, $g_3 = [\alpha = 1]$, ...). The result, shown in Fig. 23, is an incredibly clear circle in γ , which suggests that the model’s generated representation of γ lies along a circle. A simple PCA projection was not enough to find this result because the representation for γ has interference from α and β , which the EVR removes. This suggests that the models may be generating γ by using a trigonometry based algorithm like the “clock” Nanda et al. (2023a) or “pizza” Zhong et al. (2024) algorithm in late MLP layers.

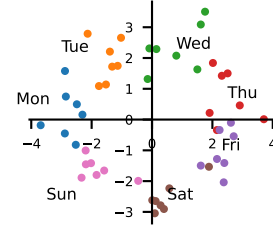


Figure 23: Top two PCA components of residual errors after EVR with one-hot in α and β . Mistral 7B `Weekdays`, layer 25, final token. Colored by γ .

K.2 MORE EXPERIMENTS WITH EVR

We now apply EVR to `Months` and `Weekdays` to break down $X_{i,l}$ completely into interpretable functions. We build a list of g_i *iteratively and greedily*. At each iteration, we perform a linear regression with the current list $g_1 \dots g_k$, visualize and interpret the residual prediction errors, and build a new function g_{k+1} representing these errors to add to the list. Once most variance is explained, we can conclude that $g_1, \dots, g_k$ constitutes the entirety of what is represented in the hidden states.